

RNPC PC & Cellphone Repair Shop

System Documentation

Current Progress: Build-a-PC Ordering, Ticketing & Repairs,
Appointment Scheduling, Edit Profile, and the Notification System

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This is a working snapshot of an actively-developed system, not a final specification.

1. Overview

RNPC Inventory & KYC is a Spring Boot (Java 21) + Thymeleaf server-rendered application for a PC and cellphone repair shop. It manages a parts catalog, a Build-a-PC configurator with checkout, walk-in repair tickets, appointment scheduling, client/KYC records, user accounts (local and Google sign-in), and an in-app notification system.

This document describes the current state of the four areas most recently built and refined: the Build-a-PC ordering flow, the walk-in Ticket and Repair Management system (including Appointment scheduling, which feeds into it), the Edit Profile feature, and the Notification system that ties the other three together.

Two roles exist: ADMIN (shop staff) and CUSTOMER. Menu visibility is adjusted per role, but only specific sensitive actions (marking an order paid, uploading a receipt, cancelling an appointment, editing a repair record) are actually enforced server-side against the signed-in account - this is called out explicitly in each section below.

2. Authentication & the Landing Page

The landing page offers a sign-in/sign-up modal with two paths:

- Continue with Google - OAuth2/OIDC sign-in. A brand-new Google account is created as CUSTOMER by default; an email can be pre-promoted to ADMIN ahead of time.
- Username / password - local accounts, checked in plaintext against the existing users table (no external identity provider involved).

Once signed in, the sidebar shows or hides menu sections (Parts, Ticket tools, KYC, Notification) based on role. Session timeouts now redirect back to the homepage instead of the standalone login page, so an idle session never strands the user on an unfamiliar screen.

3. Build a PC & Ordering (/build, /order)

3.1 Building and checking out

- /build - the customer browses eight component catalogs (CPU, GPU, Motherboard, RAM, Storage, PSU, Case, CPU Cooler) and picks one part per category (partial builds are allowed); basic socket/memory-type compatibility is checked.
- Order Now -> /order/review - shows a cart summary of the selected parts and the total price.
- Checkout -> /order/place - collects the customer's contact details, a payment method (GCash, Maribank, BPI, or RCBC, shown as radio buttons with a sample QR code for now), a payment reference number, and an optional receipt image upload. A confirmation modal appears before the order is actually submitted, to avoid accidental clicks on the 'Paid' button.

3.2 Manual payment verification

There is currently no live payment gateway wired into checkout. Xendit integration exists in the codebase but was deliberately disconnected, since a business permit for automated payment processing isn't available yet. Instead, the customer declares their own payment (bank/e-wallet + reference number, optionally a receipt photo), and an admin manually verifies it.

An order is created immediately with status `AWAITING_PAYMENT` (shown to the customer as 'Payment Verification'). The admin is notified, reviews the order from `/order` (the 'View' button opens a modal with the order's full detail, including any uploaded receipt), and clicks Mark Paid - itself gated behind its own confirmation modal. Marking an order paid records which admin verified it (their Google account ID if they've ever signed in with Google, otherwise their internal account ID) and exactly when, shown as the 'Verified by' / 'Verified at' columns visible only to admins.

The customer is notified once their order is marked paid. If they skipped the receipt upload at checkout, they can add or replace it later from an 'Upload Receipt' button in their own order list - restricted server-side to the order's own linked account. A customer may also cancel their own order while it is still awaiting payment.

4. Ticket & Repair Management (`/ticket`, `/repair`)

Two different flows feed the same Repair History: an admin creating a walk-in ticket directly, and a customer requesting an appointment (covered in the next section).

4.1 Walk-in tickets (admin-entered)

- `/ticket/create` - an admin records a client's contact details, device, brand/model, issue description, and an estimated cost.
- This looks up or creates a Client record, then creates a Repair Record with status `PENDING` and an auto-generated job order number (e.g. `JO-00007`), and prints a claim slip.
- The admin inbox gets a notification for every new ticket.

4.2 Repair History (`/repair`)

- Admins see and manage every repair record: technician, fix performed, recommendation, cost, status, warranty end date, and an optional photo.
- Customers see only their own repair records, and only once the record is actually visible to them (see 4.3) - and strictly read-only, with no edit or delete controls at all, even if the page were reached directly.

4.3 Appointment-sourced repair records

When a customer schedules an appointment (Section 5), a Repair Record is created for it automatically, but it stays hidden from that customer's own Repair History until an admin actually confirms the appointment. A walk-in ticket entered directly by an admin has no such gate and is visible immediately.

5. Appointment Scheduling (`/appointment`)

- `/appointment/create` - a customer requests a walk-in repair slot: device type, an Item Description ('specs of the laptop, desktop, or cellphone'), a History field ('what did you do before the issue occurred?'), a preferred date and time, and an issue description.
- This creates both an Appointment (status `PENDING`) and its linked, hidden Repair Record in the same step.
- The admin inbox is notified, and can review pending requests either from the `/appointment` list (each row's 'View' opens a detail modal) or from the `/appointment/calendar` month view, which shows a colour-coded chip per day.

- Confirming an appointment sets it to CONFIRMED, makes its linked repair record visible to the customer, and notifies them. Cancelling sets it to CANCELLED.
- A customer may cancel their own pending or confirmed appointment, but only while it is more than 12 hours before the scheduled time - enforced both in the UI and on the server.

6. Edit Profile (/profile/edit)

Available to every signed-in account, admin or customer alike.

- Editable fields: Name (also drives the personalised greeting on the homepage), Address, Contact Number, and Email.
- Contact Number and Email are masked by default - digits 5 through 9 of the phone number are hidden, and the middle of the email's local part is hidden (first character and last four characters remain visible, domain always shown) - each with an eye-icon toggle to reveal and edit.
- Admin accounts additionally see a read-only Employee ID field above Name. It is populated from that Google account's own unique subject ID the first time the admin signs in with Google, or falls back to their internal database ID for an admin that only ever uses a local username/password. The field is rendered disabled, so it is never editable and never even submitted with the form.

7. Notification System (/notification)

A lightweight, e-commerce-style in-app inbox. There are two audiences: a single shared inbox for every ADMIN account, and a private inbox per CUSTOMER account.

7.1 What triggers a notification

- Admin is notified when: a new order is placed, a new walk-in ticket is created, or a new appointment is requested.
- Customer is notified when: their order is marked paid, or their appointment is confirmed.
- A customer notification can only be delivered if that customer was actually signed in at the moment they placed the order or booked the appointment, since that is what links the underlying Client record to their User account. Anonymous/walk-in activity only ever reaches the admin inbox.

7.2 Reading a notification

A small unread-count badge sits next to 'Notification' in the sidebar; opening the page marks everything in that inbox as read. Clicking an individual notification does not navigate away from the list - it opens a modal populated by fetching that specific order, repair record, or appointment's own detail card (the exact same card used elsewhere in the app), so working through a long backlog never costs you your place in the list. Each fetch is still permission-checked on the server: an admin may view any record; a customer may only view a record linked to their own account.

8. Current System Flow

The diagram below traces the three main journeys - Build a PC / Order, Ticket / Repair, and Appointment - from the customer's first action through to the point each one lands in the shared Notification system, plus Edit Profile as a standalone, always-available feature.

9. Known Limitations & Not Yet Done

- Payment verification is entirely manual by design for now; the Xendit integration still exists in the codebase and can be reconnected once a business permit for automated payment processing is available.
- Menu visibility is role-based on the client side; most page routes are not hard-restricted on the server. Only the specific actions called out in this document (mark paid, upload receipt, cancel appointment, edit repair record, view someone else's notification target) currently carry an explicit server-side ownership/role check.
- There is no automated test suite yet.
- The reference PC-parts catalog was bulk-seeded from a spreadsheet; day-to-day catalog changes are made manually through the admin CRUD screens per category.

10. Appendix: Key Data Entities

Entity	Purpose
User	A login account (local or Google), its role (ADMIN/CUSTOMER), profile fields, and Google account id.
Client	A repair/order contact (name, phone, email, address); optionally linked to a User account once that person signs in while transacting.
Order / OrderItem	A Build-a-PC purchase, its line items, and its payment/verification state.
RepairRecord	A single repair job - from a walk-in ticket or a confirmed appointment - and its outcome.
Appointment	A requested walk-in repair slot and its confirm/cancel state.
Notification	An admin- or customer-facing inbox entry pointing at one specific Order, RepairRecord, or Appointment.

RNPC Inventory & KYC - Current System Flow

